



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to NAVANEETHAM AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

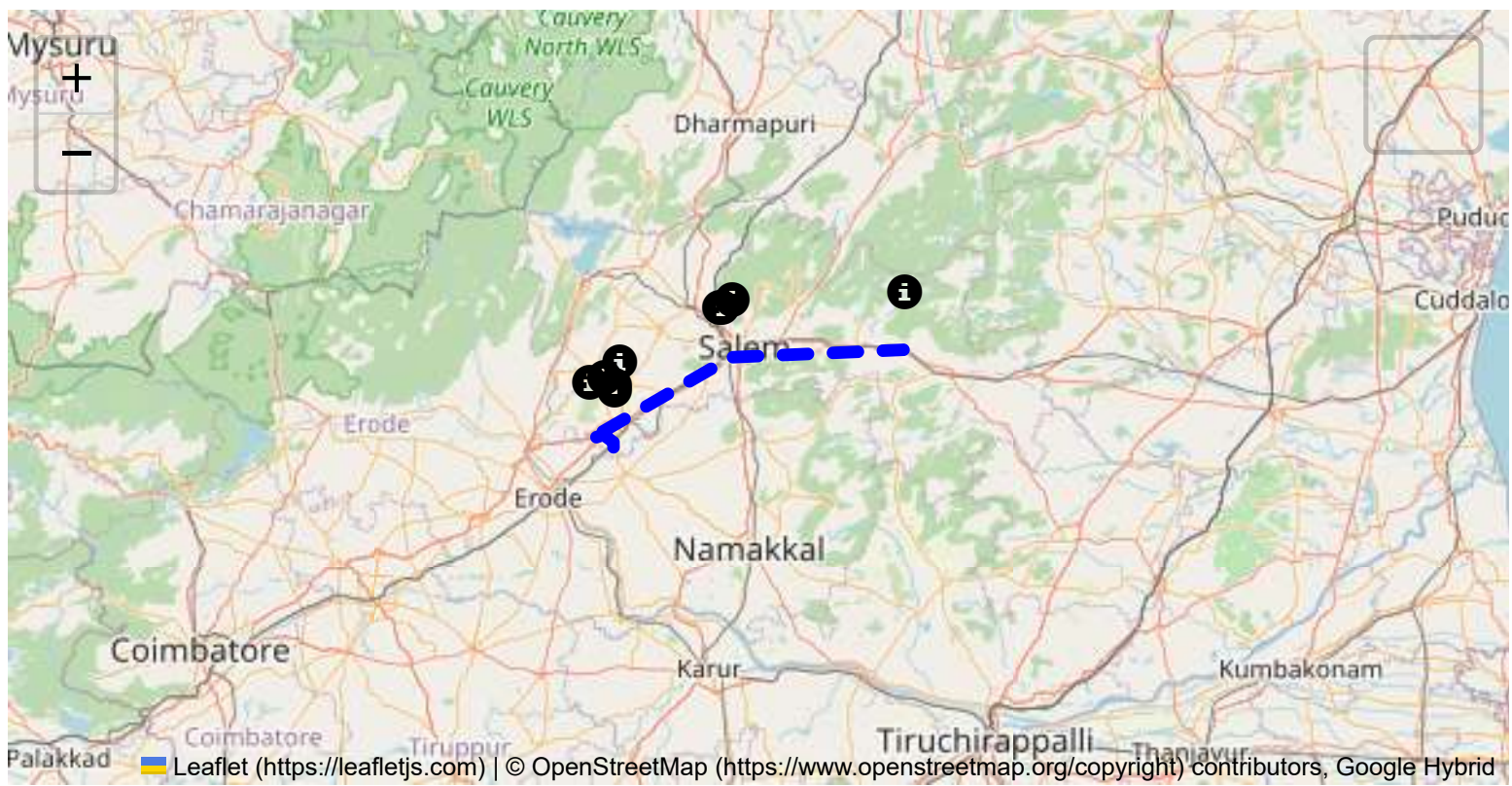
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

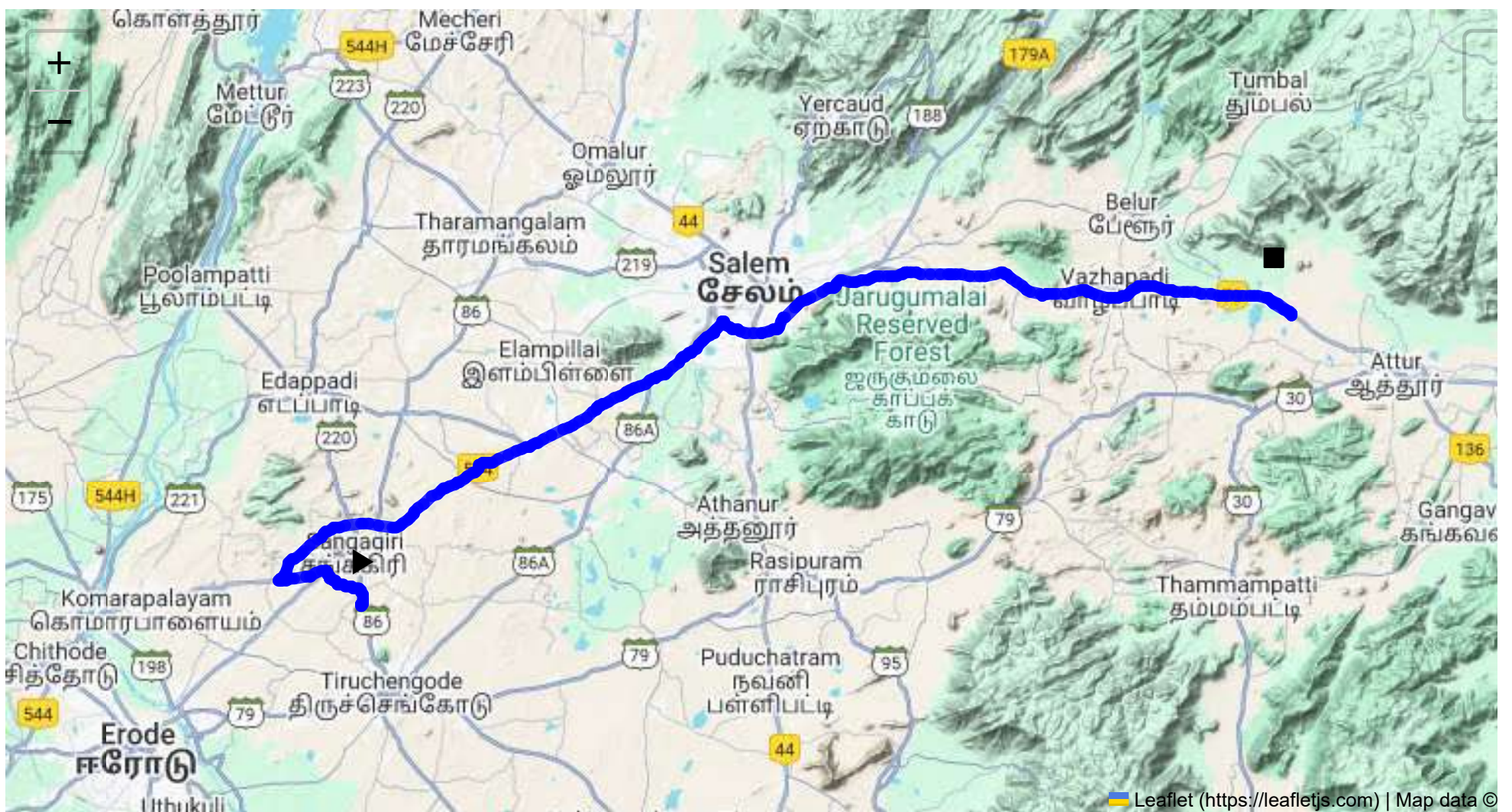
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 97.68 km
Estimated Duration: 1.8 hours
Adjusted Duration (Heavy Vehicle): 2.2 hours
Start: (11.4381, 77.8734)
End: (11.646822, 78.510895)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The specified route travels through a variety of environments, starting in Sangagiri, moving through Sankari, entering important junctions near Salem, and finally reaching Pethanaickenpalayam. It incorporates state highways, the NH544 corridor, and bypasses a significant urban area (Salem), which can impact traffic conditions and travel time.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Patterns:** Predominantly tropical climate with temperatures ranging from mild to high throughout the year.
- **Rainfall:** Higher during the monsoon period (October to December), leading to potential flooding, slippery roads, and reduced visibility.
- **Weather-Related Hazards:** Monsoon-induced flooding could cause temporary roadblocks. Heatwaves in summer months may lead to overheating of vehicles and road melting, affecting traction.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (7:00-9:00 AM) and Evening (5:00-8:00 PM) can see heavy traffic, especially in and around Salem.
- **Congestion-Prone Areas:** Salem, particularly near NH544 intersections where urban traffic flow converges, is a significant bottleneck.

4. Assessment of Road Quality and Infrastructure

- **State Highways and NH:** Generally maintained well, but periodic potholes and uneven surfaces should be expected, especially during the post-monsoon period.
- **Rural and Bypass Roads:** Some segments might be narrower with less support infrastructure, impacting larger vehicle navigation.

5. Suggestions for Alternative Routes for Emergencies

- **Alternate Routes:** In case of major congestion on NH544 or near Salem, diversions through local roads via Sankari can be considered.
- **Focused Bypass:** Utilize Salem bypasses to avoid urban congestion during emergencies.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permit Requirements:** Mandatory transport permits for hazardous materials with restrictions during peak hours to reduce risk.
- **Speed Limits and Checkpoints:** Frequent vehicular checks on highways, adherence to speed limits, and compulsory rest periods for drivers.

7. Overview of Historical Incidents

- **Recorded Incidents:** The Salem region has seen sporadic instances of truck overturns, often tied to sharp turns and road slopes.
- **Common Causes:** Most incidents involve mechanical failures and driver fatigue during long hauls.

8. Environmental Considerations and Sensitive Areas

- **Protected Areas:** Proximity to agricultural lands; special caution regarding spills and environmental contaminations.
- **Pollution Control:** Regulations on emissions and noise control, particularly when passing through residential zones near Salem.

9. Analysis of Communication Coverage

- **Coverage Strength:** Overall moderate to good cellular and network coverage throughout the route, with potential weak zones in rural segments before Salem.
- **Suggested Providers:** Major telecom operators should provide reliable service, ensure multiple network options for redundancy.

10. Estimated Emergency Response Times

- **Urban Areas (Salem):** Quick response times within 15-25 minutes due to proximity to emergency services.
- **Rural Sections:** Likely slower, ranging from 30-45 minutes depending on traffic and road conditions.

11. Overall Summary of Risk Assessment

The route traversing from CVQF+23W to JGW6+R77 holds moderate risk, primarily due to weather, traffics, and the complexity of routes near urban centers like Salem. With proper precaution, adherence to transport regulations, and awareness of traffic patterns, risks can be significantly mitigated. Emergency response structures are reasonably well-placed, though rural areas might experience delays. Planning for potential route deviations during winter months and equipping for high temperatures during summer is advised. Environmental sustainability should be observed, particularly in rural and agricultural zones.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	U-Turn	High	11.6381146, 78.5254025	10 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	Medium	11.45357, 77.85789	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
7	Turn	High	11.45352, 77.85698	15 KM/Hr
8	Turn	Medium	11.45847, 77.85140	30 KM/Hr
9	Turn	Medium	11.45898, 77.85135	30 KM/Hr
10	Turn	Medium	11.46303, 77.84945	30 KM/Hr
11	Turn	Medium	11.46314, 77.84928	30 KM/Hr
12	Turn	High	11.45542, 77.81725	15 KM/Hr
13	Turn	High	11.45634, 77.81673	15 KM/Hr
14	Turn	High	11.61233, 78.10397	15 KM/Hr
15	Turn	High	11.61216, 78.10423	15 KM/Hr
16	Turn	High	11.63300, 78.12746	15 KM/Hr
17	Blind Spot	Blind Spot	11.63803, 78.52534	10 KM/Hr
18	Turn	Medium	11.64699, 78.51104	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
17	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
18	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
19	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
20	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
21	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
22	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
23	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
24	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
25	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
26	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
31	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
33	clinic	Lavanya Clinic	11.652121, 78.4795146	30 km/h	Medium
34	hospital	Amsh Agathiyar Sidha Hospital	11.650564, 78.483542	30 km/h	Medium
35	clinic	Sudha Polyclinic	11.647825, 78.508537	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
27	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
28	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
29	school	Sri Vidhya Mandir Senior Secondary School	11.668421, 78.2509573	30 km/h	Medium
30	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
32	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.6381146, 78.5254025



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673

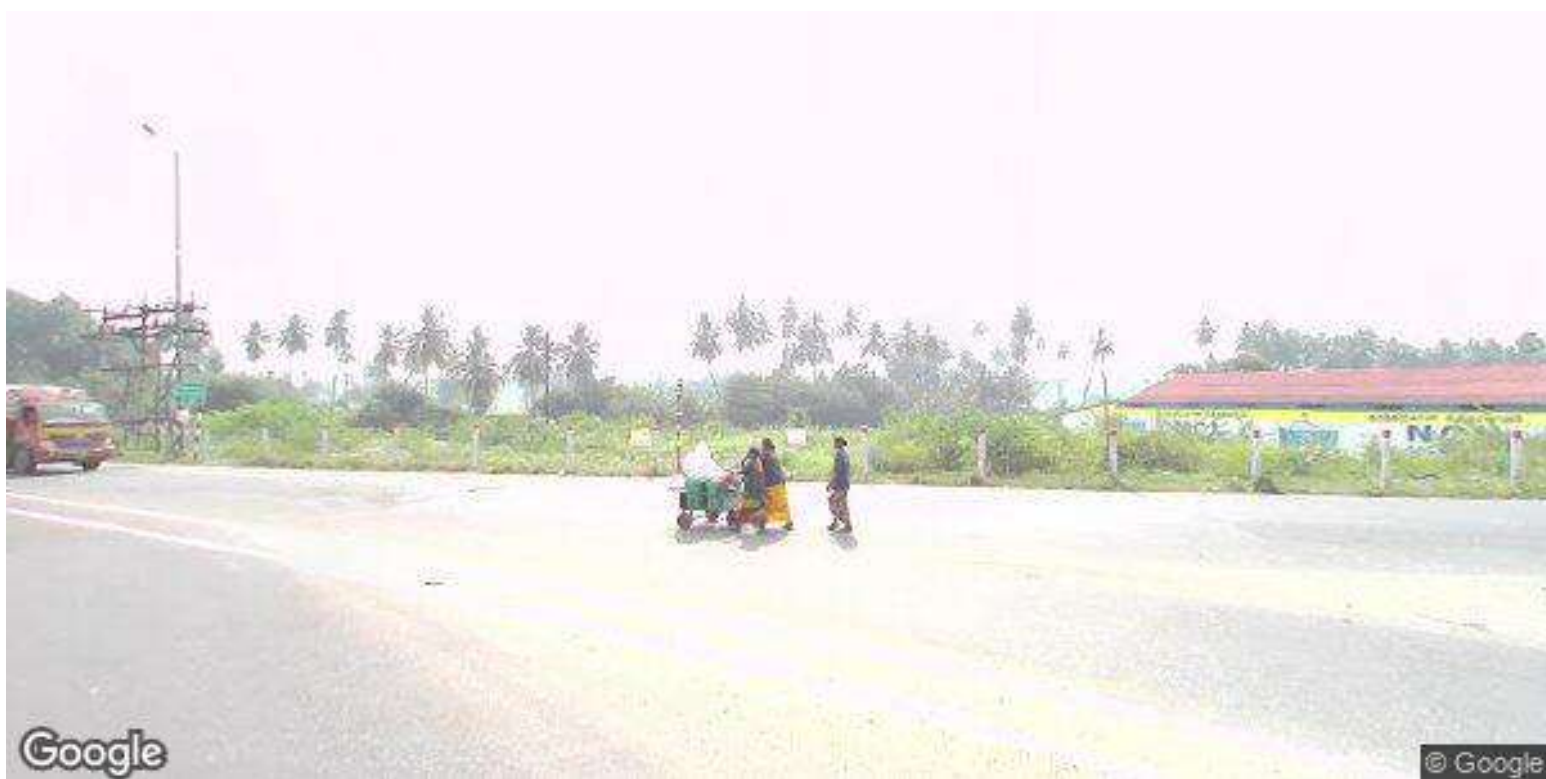


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61233, 78.10397



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.61216, 78.10423



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63300, 78.12746



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.63803, 78.52534



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64699, 78.51104
